12 Prove that the square of an odd number is always 1 more than a multiple of 4

(Total for Question 12 is 4 marks)

13 $\sqrt{5}(\sqrt{8} + \sqrt{18})$ can be written in the form $a\sqrt{10}$ where a is an integer.

Find the value of a.

×	√8	√18
√5	√40	√90

1 mark
Multiply the surds

√40 + √90 = √(4 × 10) + √(9 × 10)

1 mark
Extract the square factors

= 2√10 + 3√10 = 5√10

1 mark
The coefficient is 5

$a =$ 5

(Total for Question 13 is 3 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA